

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester Examination

University Theory Examination Nov.-Dec. 2017

PH233 Fundamentals of Packaging

Date: 10.12.2017, Sunday Time: 10.00 a.m to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any One of the following 10
A What are different functions of a package. Describe with example.
B Write a note on Testing to Assess Transport Worthiness of Package.
- Q 2** Attempt any TWO of the following 10
A What are different components of Corrugated Fibre Board. Give the examples of each component.
B 1. Which CFB Flute will you suggest
a. For Transporting Refrigerator
b. For Transporting Laptop/Telephone
2. Which adhesive will you prefer for CFB manufacturing? Why?
C Prepare a note on Structural Design of CFB.
- Q.3** Attempt the following 15
A. Suggest appropriate packaging material with rationale.
1. Paper to pack butter
2. Primary packaging material for retail packs of carbonated drinks.
3. Sealing layer in flexible packages to be manufactured for retail pack of wafer.
4. Packing milk for retail customers.
5. Wrapping material for gold ornaments.
B. Differentiate
1. Thermoplastic and Thermosetting Plastic
2. Extrusion Moulding and Blow Moulding (in plastic material manufacturing)

SECTION- II

- Q 4** Attempt any ONE of the following 10
A Write a detailed note on Aerosol, with special reference to advantages-disadvantages of the system, functioning of system as well as container for aerosols.
B Describe with help of diagram, Process of Paper Making from Pulp.
- Q 5** Attempt any TWO of the following 10
A Enlist the test to be performed on glass containers. Write significance for each test.
B Write different types of glasses, their characteristics and applications.
C What are advantages and disadvantages of glass as packaging material.
- Q.6** Do As Directed 15
1 Describe in Brief –
1. Hermetically Sealed Container
2. Annealing of Glass
2 Describe in Brief -
1. Packaging
2. Primary and Secondary Packages with Examples
3 Suggest primary packaging material to be used to pack tablet and capsule